

Item Calibration Considerations for Longitudinal Assessments in Medical Recertification

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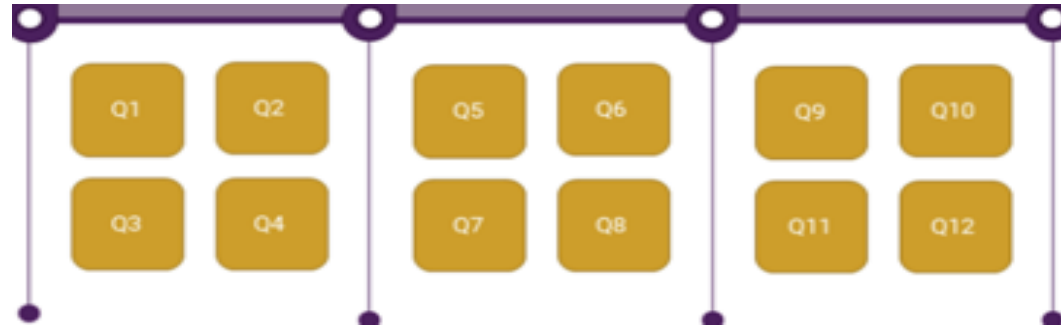
Disclosure

Nothing to Disclose

Session Learning Objectives

1. To ascertain if test questions in a longitudinal assessment significantly change in difficulty over time.
2. To evaluate risks and considerations when using item parameters for scoring in longitudinal assessments.
3. To make informed decisions about item calibration in relation to score reporting.

Overview of PANRE-LA Design



- 25 items are administered each quarter.
- Feedback information is provided after each item.
- Targeted remediation opportunities are provided after the first quarter (5 of 25 items).
- Examinees exit after 8 quarters if passing.
- Examinees can skip up to 4 quarters.
- Best 8 quarters are scored if examinee doesn't pass after the first 8 quarters.

Current Calibration Approach

Calibrations are done in the first quarter once we have enough response data for each cohort (usually around weeks 7-8 of the 10-week exam administration window).

- Data cleaning is done based on set rules, followed by Rasch analysis using WINSTEPS.
- Item parameter estimates from this analysis are used for scoring examinees throughout their LA experience.
- Items performing poorly and deemed problematic are removed from scoring.
- Quarterly cumulative scores are sent to examinees by the last business day of the quarter (approximately 14 business days after the end of exam administration).

Research Questions

1. Do item parameters significantly change at later time point during the LA experience?
2. What is the impact of scoring with item statistics from an earlier time point, given items may become easier over time?

Data Description

- Scoring data is from the current quarter of PANRE-LA (2026 Q3)
- This was Q3 for some examinees and Q7 or Q11 respectively for others.
 - Q3 examinees = 2026 cohort (N = 18,403)
 - Q7 examinees = 2025 cohort (N = 18,009)
 - Q11 examinees = 2024 cohort (N = 76)
- Item parameter estimates used for scoring are based on calibration results from the year an examinee begins PANRE-LA (cohort year).
- 491 examinees within 20 scale score points of the cut score were evaluated for impact.

Study Conditions

Condition	Description
0	Null
1	Updated item Parameter estimates at the end of Q1
2	Updated item Parameter estimates at the end of Q2

Analysis Design

- Scoring was done using item parameter estimates under different conditions.
- The impact of updating the item parameter estimates was evaluated for examinees around the cut score (± 20 scale score points).
- A total of 491 (out of 36,488) examinees fell into this category.
 - Q3 examinees = 2026 cohort (N = 410)
 - Q7 examinees = 2025 cohort (N = 67)
 - Q11 examinees = 2024 cohort (N = 14)

Results: Overall Drift

Drift in Item Difficulty from Q1 to Q2			
Year	Items in Pool	Items Flagged	% of Items Flagged
2024	711	129	18
2025	853	20	2
2026	765	17	2

NOTE: Majority of flagged items became easier.

Results: Condition 1

Increased Scores

Quarter	N	Fail	Fail -> Pass	Pass
11	9	4	1	4
7	36	4	3	29
3	339	10	107	222
Total	384	18	111	255

Reduced Scores

Quarter	N	Fail	Pass -> Fail	Pass
11	5	2	2	1
7	27	11	4	12
3	68	22	22	24
Total	100	35	28	37

NOTE: 7 examinees' scale scores did not change at all.

Results: Condition 2

Increased Scores				
Quarter	N	Fail	Fail -> Pass	Pass
11	13	2	4	7
7	67	11	7	49
3	397	49	85	263
Total	477	62	96	319

Reduced Scores				
Quarter	N	Fail	Pass - > Fail	Pass
11	1	0	0	1
7	0	0	0	0
3	7	5	0	2
Total	8	5	0	3

NOTE: 6 examinees' scale scores did not change at all.

Results

- Overall results show that it might be beneficial to update item parameter estimates, at least, at the end of Q1.
- Meanwhile, majority of examinees who were impacted by updating the item parameter estimates are in Q3 (relatively less reliable estimates based on fewer items).
- Ultimately, no examinees in this sample changed from “Pass” to “Fail” when the Q2 item parameter estimates were used.

Practical Implications of the Study

Point One

While may be beneficial to update item parameter estimates later, we also need to evaluate the feasibility of making scores available on time given strict and competing timelines.

Point Two

Since the calibration process is mostly automated after data cleaning, updating the calibration results may be relatively easy after the initial run.

Point Three

Ultimately, the final decision may involve some policy adjustments which sometimes take time to implement.



Thank You!

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